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A Simple Model for Cepheid Variability

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A simple, mathematically tractable one-zone model of Cepheid variability is constructed from the well known nonadiabatic equations governing the process. The central feature of this model is that the equation of motion is discontinuous; e. g. the equation of the compression phase differs from that of the expansion phase. (The existence of this discontinuity was deduced from astronomical observations on δ-Cephei.) The difference between the compression and expansion phases is not qualitative but merely quantitative and, mathematically speaking, it may be arbitrarily small; nevertheless, it harbors the mathematical mechanism for the limit cycle.

Necessary conditions for the existence of the limit cycle are derived, and the entire theory is applied to δ-Cephei. The predicted motion of the model preserves qualitatively and quantitatively that observed on δ-Cephei.

Key words: Cepheids — limit cycles

Introduction

In a sense, the mechanics of pulsating stars is well understood. Not only are there convincing physical descriptions of the processes leading to pulsations, but the analytical formulation of the pulsation phenomenon is also in a very satisfactory state. (See, for instance, the review article by Ledoux and Walraven, 1958). The radial pulsation of Cepheid variables is a nonadiabatic thermal and radiative process which is governed by a set of strongly nonlinear partial differential equations in which the independent variables are time and radial distance. These equations have been studied by Christy (1967) and by Cox *et al.* (1966) by means of extensive and sophisticated computation programs that would have been unthinkable before the advent of high speed computers. These studies have successfully accounted for many of the phenomena observed in the Cepheid variables.

While the *physical* processes responsible for the pulsation phenomenon are well understood, the equations governing this process are intransparent from the *mathematical* point of view. Thus, it is not possible to identify all classes of solutions of these equations, to demonstrate analytically the existence of one (or more than one) limit cycle, to decide on the existence of critical parameter values leading to bifurcations, etc. It is undoubtedly the desire to develop a mathematically more tractable theory of pulsating stars that has motivated astrophysicists, beginning perhaps with Ritter (1879) and continuing right up to the present, to develop simple models of variable stars.

One of the early efforts to discuss Cepheid variability in terms of a self-excited oscillation is due to Wesselink (1939) who qualitatively identified stellar pulsations with the newly discovered relaxation oscillations of the van der Pol equation. More recently by modifying the adiabatic pulsation equation to take account of the actual nonadiabatic character of the pulsation process, Krogdahl (1955) found an equation of the van der Pol type producing self-excited oscillations, representable as a limit cycle in the phase plane. Unfortunately, Krogdahl's limit cycle shares with Wesselink's the property of being symmetric with respect to the origin of the phase plane, while astronomical data (Schwarzschild, 1939) on Cepheids have shown that these pulsations do not possess that particular symmetry property.

The most nearly complete theory of radial pulsations in variable stars considers the star to be made up of characteristically different, contiguous spherical layers or "zones". In some, energy is absorbed from the incident radiation thus aiding the motion while, in others, energy is dissipated. For very small radial motion the energy absorption

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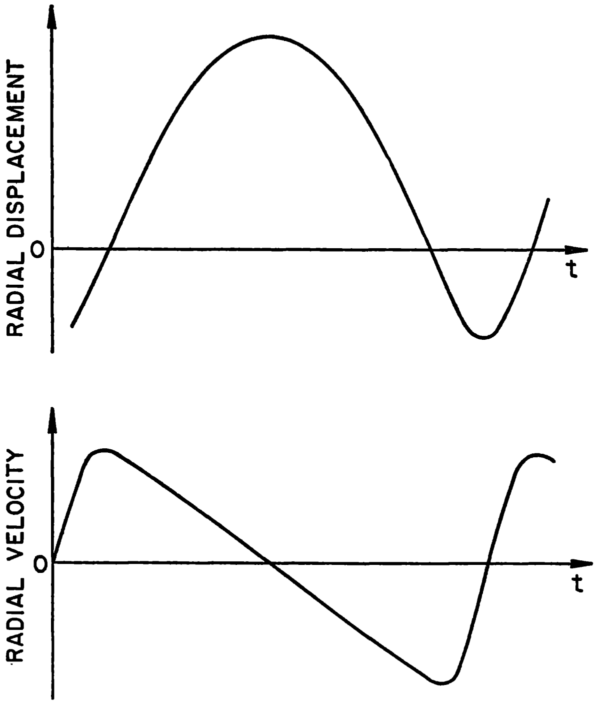

Fig. 1. Radial displacement and velocity of the surface of δ-Cephei as a function of time (after Oke, 1961)

exceeds the energy dissipation, and for large pulsations energy dissipation is preponderant. Thus, based on continuity arguments, there exists a motion such that energy absorption and dissipation balance, and periodic motion results.

Most simple models of pulsating stars are so-called one-zone models. In them, the star is considered to consist of a rigid central core which is surrounded by a homogeneous gaseous zone capable of pulsations. As this single zone is homogeneous, the only remaining independent variable is time; in this way, the governing equations become ordinary, rather than partial differential equations.

It is probable that the construction of a satisfying one-zone model leads to a third order system of ordinary differential equations. Thus Baker (1966) and Baker *et al.* (1966) dealt with a single third-order equation, while Usher and Whitney (1968) and Usher (1968) treated a coupled system of one first-order equation and one second-order equation.

Unfortunately, the existing body of knowledge of solutions of third-order nonlinear equations is small. Perhaps the earliest treatment of such a system is that of Rauch (1950) who discussed an equation which is a generalization to third order of a van der Pol equation. Rauch did find some interesting theoretical results. Baker *et al.* (1968) also found some general properties for their equation, but their most fascinating results were found by means of numerical integration. Finally, Whitney and Usher (1968) solved their system up to first-order effects by the Krylov-Bogoliubov-Mitropolski asymptotic integration method, and up to second order by a modified Poincaré-Lighthill procedure. However, general results for third-order nonlinear differential equations remain scarce, and reduction to such a system still leaves the stellar pulsation phenomenon in a mathematically unattractive form.

In this paper we propose to construct a simple one-zone model of Cepheid variables which is mathematically tractable, which is based on the known equations governing the pulsation process, and which reproduces qualitatively and quantitatively the pulsation phenomenon.

Basic Assumptions

A number of assumptions, basic to this paper, are listed below together with the reasons for their adoption.

1. The Radial Pulsation in Cepheid Variables is a Self-Excited Oscillation and thus Largely Independent of Initial Conditions

We adopt this assumption for two reasons. For one, the amplitude of pulsation of most Cepheid variables (galactic or extragalactic) is of the same order of magnitude. This is hardly to be expected in systems in which the amplitude depends on initial conditions. For another, the velocity asymmetry, discussed below and shown in Fig. 1 is a characteristic property of many self-excited oscillations, but of few other vibratory phenomena.

2. The Radial Pulsation Phenomenon in Cepheid Variables can be Modelled by a One-Zone Model

We adopted this assumption so as to arrive at a mathematically tractable theory. However, there also exist good physical grounds for this step. Consider the radial displacement and velocity of δ-Cephei, as given by Schwarzschild (1939), shown in Fig. 1. One of the striking features of the pulsation phenomenon is that the time required from maximum to minimum surface velocity is very much greater than that from minimum to maximum. Ledoux (1952) suggested that this asymmetry might be due to the fact that the outer layers of the star had, in fact, much larger motions (compared to the interior) than would be expected from uniform expansion. This conjecture was verified by Aleshin (1959) who considered a centrally condensed star in which only the outer 20 per cent of the star was vibrating. He found that the fixed core caused extreme density variations in the vibrating portion, and these could, indeed, explain the observed asymmetry.

3. The Differential Equations Governing the Compression Phase of the Pulsation Differ Quantitatively from those Governing the Expansion Phase

This assumption is based on the best astronomical data available. It is altogether central to our development and it harbors, in our opinion, the mathematical mechanism of the self-excitation. Consider the limit cycle of δ-Cephei, constructed from the data of Schwarzschild (1939) and Oke (1961) and shown in Fig. 2. The upper half-loop (in the half-plane $\dot{r} > 0$) represents the "expansion phase" of the motion, and the lower the "compression phase". These two half-loops look quite similar, leading us to believe that these two "phases" of the motion are governed by equations that are qualitatively the same (i. e. they have identical structure). This is, of course, also in accord with the known equations governing these two phases. However, the two phases of the motion occur about different equilibrium positions. At positions of equilibrium the acceleration and velocity vanish simultaneously. As one may write the acceleration in the form $\dot{r}\, d\dot{r}/dr$, the positions of equilibrium are those points on the r-axis ($\dot{r} = 0$) which lie directly above or below points on the trajectory where the tangent is parallel to the r-axis ($d\dot{r}/dr = 0$). There is one such position for each of the two half-loops; for the upper, it occurs approximately at $r/r_{\min} = 1.036$, and for the lower at $r/r_{\min} = 1.030$.

We conclude from this observation that the numerical values of the constants occurring in the governing equations are slightly different for the compression and expansion phases of the motion.

The Equation of Motion

We derive here the equation of motion of the pulsations. For the present, we are concerned only with the structure of the equations; this implies that small differences in the compression and expansion phases are not considered at this juncture.

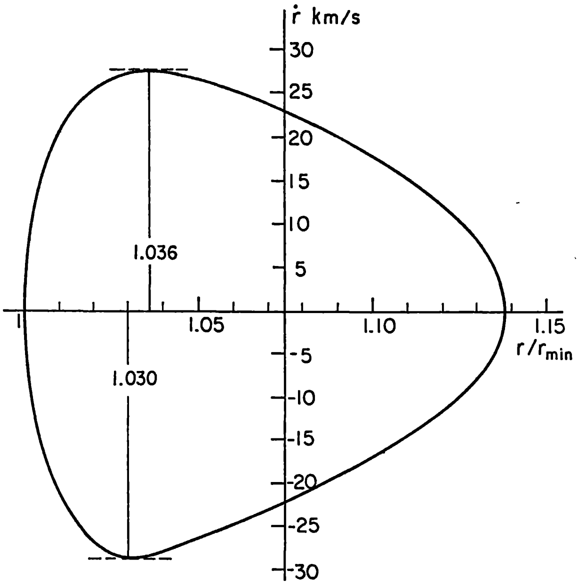

Fig. 2. The limit cycle of δ-Cephei (after Oke, 1961)

We begin with the momentum equation for the one-zone model. It will be convenient to take this in the form given by Usher and Whitney (1968); i. e.

$$m \frac{d^2 r}{dt^2} = 4\pi r^2 p + GMm/r^2 \,. \qquad (1)$$

Here, M is the mass of the solid core, $m \ll M$ is the mass of gas in the single zone surrounding the core, p is the (uniform) gas pressure, G is the gravitation constant, r is the radius of the zone, and t is the time.

We assume that there exists an equilibrium position, $r = r_0$ where $p = p_0$, and for which $d^2r/dt^2 = 0$. Equation (1) then gives

$$4\pi r_0^2 p_0 = GMm/r_0^2 \,. \qquad (2)$$

Combining (2) and (3) we may write the momentum equation in the nondimensional form

$$\frac{d^2(r/r_0)}{dt^2} = \frac{GM}{r_0^3} \left[\left(\frac{p}{p_0}\right) \left(\frac{r}{r_0}\right)^2 - \left(\frac{r_0}{r}\right)^2 \right] . \qquad (3)$$

The normal procedure for eliminating the pressure term from (3) is to differentiate (3) once with respect to time and, then, to substitute for $\dot{p}$ using the energy equation. Necessarily, this procedure leads to a third-order differential equation.

As we wish to avoid such an equation, we will consider elimination of the pressure term by integrating the energy equation rather than by differentiating the momentum equation.

If the pulsation process were adiabatic, integration of the energy equation would give

$$\frac{p}{p_0} = \left(\frac{\varrho}{\varrho_0}\right)^{\Gamma_1} \tag{4}$$

where ϱ/ϱ_0 is the density ratio and Γ_1 (assumed constant during the pulsation) is the adiabatic exponent for gas and radiation (Chandrasekhar, 1939).

When the pulsation is nonadiabatic (or when Γ_1 varies during the oscillation) the simple relation (4) no longer holds. In this case, the energy equation integrates to the form (Ledoux and Walraven, 1958).

$$\frac{p}{p_0} = K(r_0, t)\left(\frac{\varrho}{\varrho_0}\right)^{\Gamma_1}. \tag{5}$$

As we wish to account for the nonadiabatic character of the phenomenon, we do not adopt (4). Instead, we make use of the fact that, on any open interval bounded by successive maxima and minima of r, the solution $r(t)$ has an inverse $t(r)$. Thus, (5) is of the form

$$\frac{p}{p_0} = K(r_0, t(r))\left(\frac{\varrho}{\varrho_0}\right)^{\Gamma_1}, \tag{6}$$

and the function $K(r_0, r)$ must properly reflect the nonadiabatic process. We shall write

$$\frac{p}{p_0} = \left(\frac{\varrho}{\varrho_0}\right)^{\Gamma_1} h(1 - r/r_0) \tag{7}$$

where $h \neq$ *constant* is a function (of $(1 - r/r_0)$) which incorporates the nonadiabatic character of the pulsation mechanism of the single zone, and whose structure will be discussed in detail later on. Note that in the adiabatic case $h(1 - r/r_0) \equiv 1$.

A relation between density and radius is provided by the continuity equation. For the one-zone model this is (Usher and Whitney, 1968)

$$\frac{\varrho}{\varrho_0} = \frac{n_0^3 - 1}{(r/r_0)^3\, n_0^3 - 1} \tag{8}$$

when $n_0 = r_0/r_c$, and r_c is the radius of the fixed core.

For the small range of r/r_0 occurring in Cepheid variables (approximately $0.95 \leqq r/r_0 \leqq 1.1$) one may assume that there exists a real constant $m > 0$ such that

$$(\varrho/\varrho_0) = (r_0/r)^m . \tag{9}$$

By combining (8) and (9) one sees that for a finite core ($n_0 > 1$), m will be much greater than 3[1]). This is the result found by Aleshin (1959).

[1]) In the case of uniform expansion of a homogeneous, compressible sphere, $m = 3$.

The substitution of (7) and (9) in (3) gives the differential equation

$$\frac{d^2(r/r_0)}{dt^2} = \frac{GM}{r_0^3}\left[\left(\frac{r}{r_0}\right)^{2 - m\Gamma_1} h(1 - r/r_0) - (r_0/r)^2\right]. \tag{10}$$

Finally, if one puts

$$\frac{r}{r_0} = Q, \quad t\frac{GM}{r_0^3} = \tau, \quad m\,\Gamma_1 - 2 = n,$$

(10) becomes

$$\frac{d^2Q}{Q\tau^2} = \frac{h(1-Q)}{Q^n} - \frac{1}{Q^2}. \tag{11}$$

We regard (11) as the equation which governs the compression (or the expansion) phase of the stellar pulsation. The nature of the function $h(1 - Q)$ and the magnitude of n in that equation will be determined from astronomical data.

One conclusion drawn from astronomical observation (and already discussed earlier) is that the equations governing expansion and compression are alike, but not identical. Thus we write

$$\left.\begin{aligned} \frac{d^2Q}{d\tau^2} &= \frac{h_e(1 - Q)}{Q^n} - \frac{1}{Q^2}, \quad \dot{Q} > 0, \\ \frac{d^2Q}{d\tau^2} &= \frac{h_c(1 - Q)}{Q^n} - \frac{1}{Q^2}, \quad \dot{Q} < 0, \end{aligned}\right\} \tag{12}$$

and we require, as in all hamiltonian systems, that Q and $\dot{Q}$ be everywhere continuous functions of time, including the instants of rest when $\dot{Q} = 0$. The two Eqs. (12) may be combined into the single equation

$$\frac{d^2Q}{d\tau^2} = \frac{1}{2}\left(\frac{h_e - h_c}{Q^n}\right) sgn\, \dot{Q} + \frac{1}{2}\left(\frac{h_e + h_c}{Q^n} - \frac{2}{Q^2}\right) \tag{13}$$

where $sgn\,\dot{Q} = \pm 1$ according as $\dot{Q} > 0$ or $\dot{Q} < 0$. However, there is no particular advantage in dealing with (13) in place of (12).

The Exciter Mechanism

We shall suppose that the functions h_e and h_c are analytic near $Q = 1$. Thus, we may write

$$\left.\begin{aligned} h_e(1 - Q) &= H_{e0} + \sum_{i=1,2,\ldots} H_{ei}(1 - Q)^i, \\ h_c(1 - Q) &= H_{c0} + \sum_{i=1,2,\ldots} H_{ci}(1 - Q)^i, \end{aligned}\right\} \tag{14}$$

where the H_{ej} and H_{cj}, ($j = 0, 1, 2, \ldots$) are constants whose numerical values will be determined from astronomical observations.

It is clear from (12) and (14) that there exists a sufficiently small neighborhood about $Q = 1$ in which the motion is governed by

$$\left.\begin{aligned} \frac{d^2Q}{d\tau^2} &= \frac{H_{e0}}{Q^n} - \frac{1}{Q^2}\,, \quad \dot{Q} > 0\,, \\ \frac{d^2Q}{d\tau^2} &= \frac{H_{c0}}{Q^n} - \frac{1}{Q^2}\,, \quad \dot{Q} < 0\,. \end{aligned}\right\} \quad (15)$$

Occasionally, we shall regard (15) as the equations of rectilinear motion of a unit mass subjected to forces given by the right-hand sides of (15).

We now consider the two constants H_{e0} and H_{c0}. They must be so chosen as to produce unstable oscillations about $Q = 1$.

It can easily be shown that the first of (15) has an equilibrium position at $Q = H_{e0}^{1/(n-2)}$, and the second at $Q = H_{c0}^{1/(n-2)}$. Moreover, it is known from Aleshin's work that $n > 3$. (Actually, we have only shown that $m > 3$. However, in a typical one-zone model for which the thickness of the zone is less than $0.2\, r_0$, one finds from (8) and (9) that $m \simeq 8$; then, since $\Gamma_1 > 1$ for all real gases, ionized or not, $n > 3$ because $n = m\,\Gamma_1 - 2$.) In addition, our later discussion of the energy of the motion shows that $n \geqq 3$ is a necessary condition for the motion of (12) to be oscillatory. Therefore, we impose $n \geqq 3$.

From astronomical observations of δ-Cephei one has

$$H_{e0} > 1 > H_{c0} \qquad (16)$$

as may readily be verified from Fig. 2. We shall now show that, when the constants in (15) satisfy the inequality (16), and when $n \geqq 3$, the solutions to (15) are unstable oscillations.

Let Q_1 denote the "inner" amplitude of Q (i. e. the rest value of Q at the end of the compression phase), and Q_2 the "outer" amplitude of Q (i. e. the rest value of Q at the end of the expansion phase). Then these two quantities satisfy necessarily

$$Q_2 > 1 > Q_1\,. \qquad (17)$$

In order to demonstrate that the motion is an oscillation of increasing amplitude we shall assume, to the contrary, that the motion is periodic. In that case, the total work done in one complete cycle by the forces H_{e0}/Q^n and H_{c0}/Q^n must vanish. However, we shall show this work is, in fact, positive, thus producing an oscillation of increasing amplitude.

Let chronologically neighboring rest points be denoted by $Q_0, Q_1, Q_2, \ldots$ with $Q_0 > 1$; then, it follows that $Q_0, Q_2, \ldots > 1$ and $Q_1, Q_3, \ldots < 1$. If the motion is *periodic*, $Q_0 = Q_2 = \ldots$, and $Q_1 = Q_3 = \ldots$, and the work done by the forces

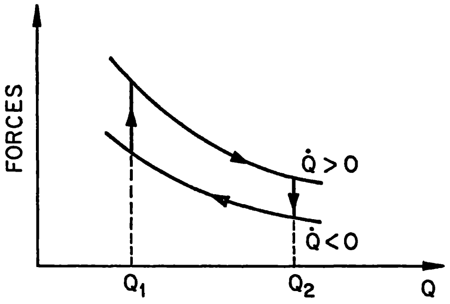

Fig. 3. Negative hysteresis loop for small motions

H_{e0}/Q^n and H_{c0}/Q^n is

$$\begin{aligned} W &= \int_{Q_1}^{Q_2} \frac{H_{e0}}{Q^n}\, dQ - \int_{Q_1}^{Q_2} \frac{H_{c0}}{Q^n}\, dQ \\ &= \frac{H_{e0} - H_{c0}}{n-1} \left(\frac{1}{Q_1^{n-1}} - \frac{1}{Q_2^{n-1}} \right). \end{aligned} \qquad (18)$$

But $H_{e0} \neq H_{c0}$, and $Q_1 \neq Q_2$. Hence, $W \neq 0$ which contradicts the supposition that the motion is periodic. In fact, $H_{e0} > H_{c0}$, $Q_2 > Q_1$, and $n \geqq 3$. It follows from these inequalities that the forces in question do positive work on the motion. We conclude that the motion is oscillatory because $n \geqq 3$, and the amplitude increases because W in (18) is positive.

This result can be readily interpreted in terms of a "negative hysteresis loop" closely related to a pressure-volume diagram. In Fig. 3, the right-hand sides of (15) are plotted as two curves. The curve for $\dot{Q} > 0$ lies everywhere above the one for $\dot{Q} < 0$ because $H_{e0} > H_{c0}$. Let Q_1 and Q_2 be two rest points satisfying the inequality (17). Then, the two curves and the lines $Q = Q_1$, and $Q = Q_2$ in Fig. 3 form a loop which is traversed in the direction determined by the sign of $\dot{Q}$ and indicated by the arrows in Fig. 3. When the loop is traversed in the counter-clockwise direction energy is removed, and the loop is then commonly called a hysteresis loop. Here, the loop is traversed in the clockwise direction leading to the terminology "negative hysteresis loop". From the relation between pressure and force on the one hand, and volume and radial displacement on the other, it is evident how Fig. 3 is related to the pressure-volume diagram.

It is instructive to consider the system (15) in the phase plane; i. e.,

$$\left.\begin{aligned} \frac{d\dot{Q}}{dQ} &= \frac{(H_{e0} - Q^{n-2})/Q^n}{\dot{Q}}\,, \quad \dot{Q} > 0\,, \\ \frac{d\dot{Q}}{dQ} &= \frac{(H_{c0} - Q^{n-2})/Q^n}{\dot{Q}}\,, \quad \dot{Q} < 0\,. \end{aligned}\right\} \quad (19)$$

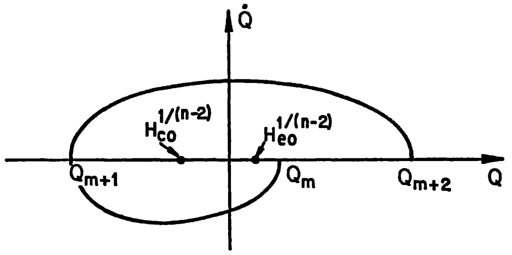

Fig. 4. Phase plane trajectory for small motions

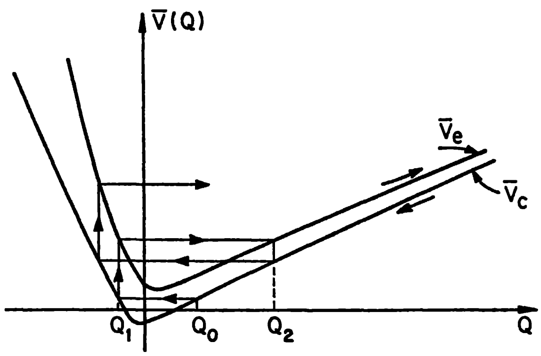

Fig. 5. Potential energy curves ($\bar{V}_e$ for expansion and $\bar{V}_c$ for compression) for small motion

The first of (19) has a center at ($Q = H_{e0}^{1/(n-2)}$, $\dot{Q} = 0$), and the second has a center at ($Q = H_{c0}^{1/(n-2)}$, $\dot{Q} = 0$). Therefore, the integral curves of the first of (19) are half-ovals in the upper half of the phase plane, centered on $(H_{e0}^{1/(n-2)}, 0)$, and those of the second are half-ovals in the lower half, centered on $(H_{c0}^{1/(n-2)}, 0)$. The continuity conditions on Q and $\dot{Q}$ have the consequence that every trajectory is continuous across the Q-axis. Moreover, the slopes $d\dot{Q}/dQ$ of both of (19) are infinite at the Q-axis intercepts; therefore, the trajectories are in fact continuous and smooth everywhere. It has already been shown that the amplitude of the motion increases with time. Therefore, a trajectory of the motion looks as shown in Fig. 4, and time increases in the clockwise direction.

For $n = 3$, the rate of amplitude growth is readily computed. Let $Q_m > 1$ be a rest point, and let $Q_{m+1}, Q_{m+2}, \ldots$, be chronologically successive rest points. Then, integrating the first of (19) from Q_{m+1} to Q_{m+2}, and the second from Q_m to Q_{m+1}, one finds the two equations

$$\frac{H_{e0}}{2Q_{m+2}^2} - \frac{1}{Q_{m+2}} = \frac{H_{e0}}{2Q_{m+1}^2} - \frac{1}{Q_{m+1}},$$

$$\frac{H_{c0}}{2Q_{m+1}^2} - \frac{1}{Q_{m+1}} = \frac{H_{c0}}{2Q_m^2} - \frac{1}{Q_m}.$$

Continuity at the Q-axis intercept implies that the Q_{m+1} in these two equations are the same. Eliminating that quantity between them results in

$$\frac{1}{Q_{m+2}} - \frac{1}{Q_m} = 2\left(\frac{1}{H_{e0}} - \frac{1}{H_{c0}}\right)$$

where Q_m and Q_{m+2} are spatially adjacent rest points which are chronologically separated by Q_{m+1} as shown in Fig. 4. Since $H_{e0} > H_{c0}$, the above equation shows again that $Q_{m+2} > Q_m$.

We may write (15) in the form

$$\left.\begin{aligned} \frac{d^2Q}{d\tau^2} &= -\frac{d\bar{V}_e}{dQ}, \quad \dot{Q} > 0, \\ \frac{d^2Q}{d\tau^2} &= -\frac{d\bar{V}_c}{dQ}, \quad \dot{Q} < 0, \end{aligned}\right\} \quad (20)$$

where

$$\left.\begin{aligned} \bar{V}_e &= \frac{H_{e0}}{(n-1)\,Q^{n-1}} - \frac{1}{Q}, \\ \bar{V}_c &= \frac{H_{c0}}{(n-1)\,Q^{n-1}} - \frac{1}{Q}. \end{aligned}\right\} \quad (21)$$

Then, if we define $T = \frac{1}{2}\dot{Q}^2$, (20) admits the energy integrals

$$T + \bar{V}_e = \text{const.}, \quad \dot{Q} > 0,$$

$$T + \bar{V}_c = \text{const.}, \quad \dot{Q} < 0.$$

Having regarded (15) as the equations of motion of a unit mass, one would conclude in the terminology of analytical mechanics that energy is conserved while the velocity is positive (during the expansion phase) and while it is negative (during the compression phase), but a discontinuous change of energy occurs at the transitions from one to the other and, here, this change is a net gain during one complete oscillation.

(If an energy integral of our equations of motion exists on any domain, we shall say that energy is conserved on that domain. We use this terminology whether the model under consideration is adiabatic or nonadiabatic. If this terminology seems strange it should be remembered that our nonadiabatic model is only piecewise conservative, but globally nonconservative. Moreover, the discontinuous energy change is merely an idealization of a continuous process with isolated, locally steep gradients.)

It is instructive to examine the potential energy curves $\bar{V}_e(Q)$ and $\bar{V}_c(Q)$ in more detail. It follows from (21) that $\bar{V}_e$ has a minimum at $Q = H_{e0}^{1/(n-2)}$, and $\bar{V}_c$ has a minimum at $Q = H_{c0}^{1/(n-2)}$. Also,

$$\bar{V}_e - \bar{V}_c = \frac{H_{e0} - H_{c0}}{(n-1)\, Q^{(n-1)}} \,. \tag{22}$$

Therefore, the $\bar{V}_e$-curves lie everywhere above the $\bar{V}_c$-curve and the difference (22) increases as $Q < 1$ decreases, and it decreases as $Q > 1$ increases. Curves having these properties are shown in Fig. 5.

Suppose the motion begins at a rest point $Q_0 > 1$; i. e., Q_0 is the (nondimensional) radius at the end of the expansion phase. Then, the next rest point Q_1 occurs at the end of the compression phase, as shown in Fig. 4. Since the system now enters on an expansion phase, the next rest point is found in accordance with the construction in Fig. 5. From here on, the entire process repeats itself. It is seen that the amplitude of the motion grows, and there is a net increase in energy. In fact, there is an energy gain in the transition from compression to expansion, and a loss in the transition from expansion to compression, but the energy gain exceeds the loss.

The Limit Cycle

From the discussion of the exciter mechanism it is seen that, when the argument $(1 - Q)$ of the functions h_e and h_c is sufficiently small, the motion is unstable, the amplitude increases with time, and $(1 - Q)$ does not remain small. Thus, we must consider the differential Eq. (12) with

$$\left.\begin{aligned} h_e &= H_{e0} + H_{e1}(1-Q) + H_{e2}(1-Q)^2 + \dots, \\ h_c &= H_{c0} + H_{c1}(1-Q) + H_{c2}(1-Q)^2 + \dots, \end{aligned}\right\} \tag{23}$$

where H_{e0} and H_{c0} satisfy the inequality (16) and the H_{ei}, H_{ci}, $(i = 1, 2, \dots)$ are constants. They must have values such that the system (12) has a stable limit cycle whose shape resembles the observed limit cycle of Fig. 2.

To establish the values of the constants in (23) we observe that the function $K(r_0, t)$ in (5) may be written as (Ledoux and Walraven, 1958)

$$K(r_0, t) = \exp\left[-\int_0^t \dot{\Gamma}_1 \log(\varrho/\varrho_0)\, dt\right].$$

Evidently, this quantity is necessarily positive for every bounded t; hence, $K(r_0, t)$ cannot change sign. Then, it follows from (6) and (7) that we should

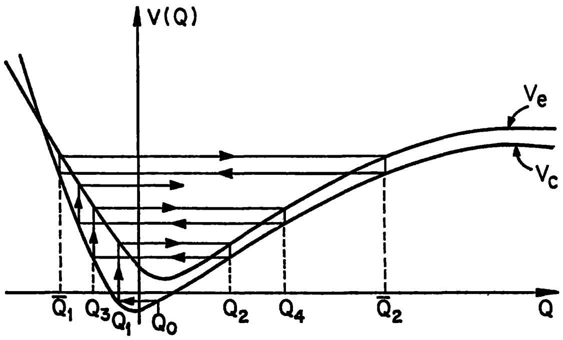

Fig. 6. Potential energy curves for large motion

choose for the $h(1-Q)$ a form which also does not change sign for any Q. Moreover, $|1 - Q|$ is small. Combining these two observations with (23) leads to

$$\left.\begin{aligned} h_e &= H_{e0} + H_{e2}(1-Q)^2, \\ h_c &= H_{c0} + H_{c2}(1-Q)^2. \end{aligned}\right\} \tag{24}$$

It follows that the equations considered from this point on may be written as

$$\left.\begin{aligned} \frac{d^2Q}{d\tau^2} &= -\frac{dV_e}{dQ}, \quad \dot{Q} > 0, \\ \frac{d^2Q}{d\tau^2} &= -\frac{dV_c}{dQ}, \quad \dot{Q} < 0, \end{aligned}\right\} \tag{25}$$

with

$$V_e = \int \left[\frac{H_{e0} + H_{e2}(1-Q)^2}{Q^n} - \frac{1}{Q^2}\right] dQ$$

and a similar expression for V_c; i. e.

$$\left.\begin{aligned} V_e &= \frac{H_{e0} + H_{e2}}{(n-1)\, Q^{n-1}} - \frac{2H_{e2}}{(n-2)\, Q^{n-2}} + \frac{H_{e2}}{(n-3)\, Q^{n-3}} - \frac{1}{Q}, \\ V_c &= \frac{H_{c0} + H_{c2}}{(n-1)\, Q^{n-1}} - \frac{2H_{c2}}{(n-2)\, Q^{n-2}} + \frac{H_{c2}}{(n-3)\, Q^{n-3}} - \frac{1}{Q}. \end{aligned}\right\} \tag{26}$$

We shall show that

$$H_{c2} - H_{e2} > 0 \tag{27}$$

is a necessary condition for the existence of a limit cycle. However, the concepts and computations needed to demonstrate this result are involved. Therefore, we postpone this demonstration until later, and we begin with a descriptive explanation of the mechanism of the limit cycle.

In Fig. 6 we show V_e and V_c-curves as functions of Q where the constants were so chosen as to satisfy the above conditions for the existence of a limit cycle. As in Fig. 5, if Q_0 is a rest point at the end of an expansion phase, and if $Q_0 - 1$ is small, the motion builds up. However, when (27) is satisfied, there will exist a rest point $\bar{Q}_1$ and a corresponding next rest point $\bar{Q}_2$ such that the energy gain $V_e(\bar{Q}_1) - V_c(\bar{Q}_1)$ at the end of a compression phase is exactly equal to the energy loss $V_c(\bar{Q}_2) - V_e(\bar{Q}_2)$ at the end of the expansion phase. Therefore, the oscillation between the rest points $\bar{Q}_1$ and $\bar{Q}_2$ is a periodic motion and is, in fact, the limit cycle.

Mathematically, the limit cycle corresponds to a fixed point of a certain mapping of the positive, real line into itself. To see this, consider the rest point Q_0 in Fig. 6; i. e. $\dot{Q}(Q_0) = 0$; the next rest point Q_1 is found from $\dot{Q}(Q_1) = 0$. Thus, there exists a transformation T_1 defined by the energy integral of the second of (25) which maps Q_0 into Q_1. In a similar way, the rest point Q_2 is the image of Q_1 from a mapping T_2, defined by the energy integral of the first of (25). Then the mapping

$$T = T_2 \circ T_1 : \quad R^+ \to R^+$$

where the circle indicates the composition of the two mappings, and R^+ is the positive real line, is in fact a transformation

$$T: Q_0 \to Q_2$$

which maps Q_0 into Q_2. This last equation is frequently written by physical scientists as

$$Q_2 = Q_2\,(Q_0)\,.$$

If there exists a $Q_0 = Q_0^*$ which is mapped into itself by T; i. e.

$$Q_2\,(Q_0^*) = Q_0^*\,,$$

Q_0^* is a fixed point of the transformation T. One sees easily from Fig. 6, in what way the limit cycle is identified with the fixed point of the transformation

$$T: Q_0 \to Q_2 \to Q_4 \to \ldots \to \bar{Q}_2 \to \bar{Q}_2\,,$$

but it turns out that it is difficult to compute the explicit form of the composition $T_2 \circ T_1$. Therefore, the existence of the limit cycle will be identified with the fixed point of a different, but related, transformation T'.

Consider the equation of motion of the compression phase

$$\frac{d^2Q}{d\tau^2} = \frac{H_{c0} + H_{c2}\,(1-Q)^2}{Q^n} - \frac{1}{Q^2}\,. \tag{28}$$

This is, in fact, the second of (25) combined with the second (26). Integrating this equation once between chronologically neighboring rest points Q_0 and Q_1 gives

$$0 = -\frac{H_{c0} + H_{c2}}{n-1}\left[\frac{1}{Q_1^{n-1}} - \frac{1}{Q_0^{n-1}}\right] + \frac{2H_{c2}}{n-2}\left[\frac{1}{Q_1^{n-2}} - \frac{1}{Q_0^{n-1}}\right] - \frac{H_{c2}}{n-3}\left[\frac{1}{Q_1^{n-3}} - \frac{1}{Q_0^{n-3}}\right] + \left[\frac{1}{Q_1} - \frac{1}{Q_0}\right]. \tag{29}$$

Equation (29) is the implicit form of the transformation T_1.

We now define new variables

$$S_0 = Q_1/Q_0\,,\; P_1 = 1/Q_1\,, \tag{30}$$

and we note that, while the motion builds up, $S_0 < 1$, and in all cases $P_1 > 1$. In terms of the new variables, (29) becomes

$$0 = -\frac{H_{c0} + H_{c2}}{n-1} P_1^{n-1}\,(1 - S_0^{n-1}) + \frac{2H_{c2}}{n-2} P_1^{n-2}\,(1 - S_0^{n-2}) - \frac{H_{c2}}{n-3} P_1^{n-3}\,(1 - S_0^{n-3}) + P_1\,(1 - S_0)\,. \tag{31}$$

This is the implicit form of a transformation T_1', which maps P_1 into S_0.

If one forms the derivative dS_0/dP_1 from (31) and utilizes the facts that $dV_c/dQ|_{Q_0} > 0$, that $dV_c/dQ|_{Q_1} < 0$, and that $P_1 S_0 > 0$, one can show after a lengthy calculation not reproduced here (Rudd, 1969) that

$$\frac{dS_0}{dP_1} < 0\,. \tag{32}$$

To construct the mapping $S_0(P_1)$, we divide (31) by P_1^{n-1}, we remove the trivial solution $S_0 = 1$ (no motion), and we let

$$\frac{H_{c0} + H_{c2}}{(n-1)\,H_{c2}} = a\,. \tag{33}$$

This gives the implicit relation

$$0 = -\frac{1 - S_0^{n-1}}{1 - S_0} + \frac{2}{a\,(n-2)\,P_1}\,\frac{(1 - S_0^{n-2})}{(1 - S_0)} - \frac{1}{a\,(n-3)\,P_1^2}\,\frac{(1 - S_0^{n-3})}{(1 - S_0)} + \frac{1}{a\,H_{c2}\,P_1^{n-2}}\,. \tag{34}$$

This equation defines a curve in the $S_0 P_1$-plane whose slope is negative in view of (32).

If one considers instead of (28) the equation of the expansion phase, integrates it between Q_1 and

Q_0, and introduces into it

$$S_2 = Q_1/Q_2\,, \quad P_1 = 1/Q_1\,, \tag{35}$$

one finds, analogous to (31),

$$\begin{aligned} 0 = &-\frac{H_{e0}+H_{e2}}{n-1} P_1^{n-1}(1-S_2^{n-1}) \\ &+\frac{2H_{e2}}{n-2} P_1^{n-2}(1-S_2^{n-2}) \\ &-\frac{H_{e2}}{n-3} P_1^{n-3}(1-S_2^{n-3}) + P_1(1-S_2)\,, \end{aligned} \tag{36}$$

and one can show that, here also,

$$\frac{dS_2}{dP_1} < 0\,. \tag{37}$$

As an example, the curves $S_0(P_2)$ and $S_2(P_2)$ in Fig. 7 are plotted for $n=3$, $H_{e0}=1.01$, $H_{e2} = 0.178$, $H_{c0}=0.99$, $H_{c2}=2.000$.

Consider some $P_1 = P_1^0$, as shown in Fig. 7. To it there correspond the two values

$$S_0^0 = S_0(P_1^0)\,, \quad S_2^0 = S_2(P_1^0)\,.$$

The elimination of P_1^0 between these two defines the mapping

$$T'\colon S_0 \to S_2$$

or

$$S_2 = S_2(S_0)\,. \tag{38}$$

Now, we have shown that a periodic solution exists where the mapping T has a fixed point; i.e., where $Q_2/Q_0 = 1$. But, when $Q_2/Q_0=1$, it follows from (30) and (35) that $S_0/S_2 = 1$ as well. In other words, a periodic solution is also identified with a fixed point of the mapping T'. In the example of Fig. 7, there exists a value P_1, for which the mapping T' has a fixed point.

If there exists a fixed point when $P_1 = \bar{P}_1$, one has

$$\bar{S}_0 = \bar{S}_2 = \bar{S} \tag{39}$$

where

$$\bar{S}_0 = S_0(\bar{P}_1)\,, \quad \bar{S}_2 = S_2(\bar{P}_1)\,.$$

Substituting $P_1 = \bar{P}_1$ and (39) into both (34) and the corresponding equation for the expansion phase, and combining these two equations, one finds after some manipulations

$$\bar{P}_1^{n-2}\frac{(1-\bar{S}^{n-1})}{(1-\bar{S})} = \frac{(n-1)(H_{c2}-H_{e2})}{H_{e0}H_{c2}-H_{c0}H_{e2}}\,. \tag{40}$$

Then, a necessary and sufficient condition for the existence of a fixed point of T' is that there exists a real value $\bar{P}_1$ for which (40) is satisfied.

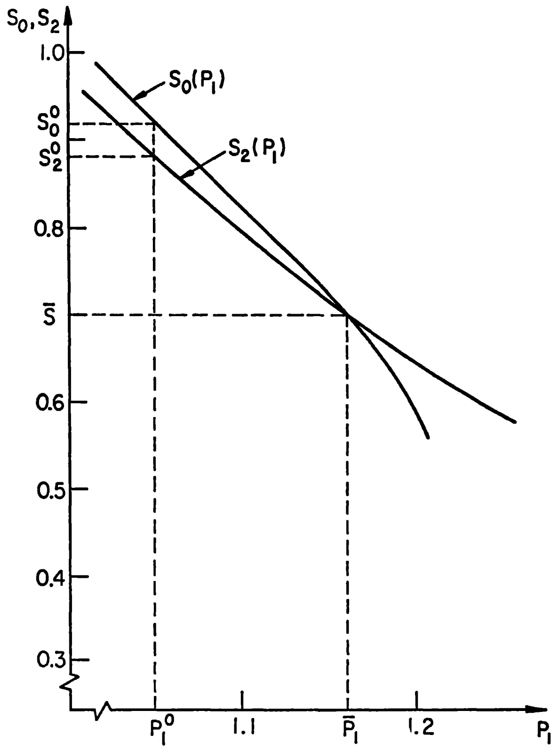

Fig. 7. The mapping from S_0 to S_2

We now consider the left-hand side of (40)

$$F(P_1) = \frac{P_1^{n-2}[1-S(P_1)^{n-1}]}{1-S(P_1)}\,, \tag{41}$$

and we note that $F(P_1)$ is positive for all possible $P_1 > 0$ because $0 < S < 1$. Thus, a necessary condition for (40) to hold is that the constants on the right-hand side be such that the fraction

$$\frac{(n-1)(H_{c2}-H_{e2})}{H_{e0}H_{c2}-H_{c0}H_{e2}} > 0\,. \tag{42}$$

But, when H_{e0} and H_{c0} satisfy (16) and when $n \geqq 3$, the inequality (42) is satisfied whenever

$$H_{c2} - H_{e2} > 0 \tag{43}$$

which was to be shown.

In this manner, the structure and relative value of the functions $h(1-Q)$ is completely determined, in part from astronomical observations, in part from purely mathematical considerations mentioned in connection with (24), and in part from the requirement that the star oscillates in a limit cycle. However, there is also physical evidence to support our

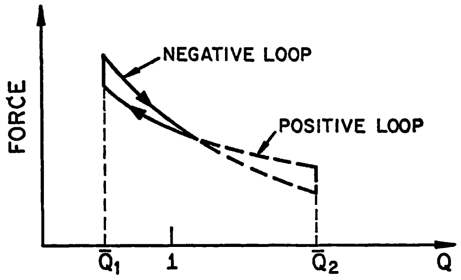

Fig. 8. Hysteresis loop for periodic motion

theory[2]). Thus, one may write for the function in (5)

$$K(r_0, t) = \exp\left[-\int_0^t (\Gamma_3 - 1)\frac{\varrho}{p}\frac{dL}{dm}\,dt\right]$$

with dL/dm positive in the expansion phase and negative in the compression phase. This certainly agrees qualitatively with the laws established by us.

In Fig. 3 we examined the hysteresis loop for motion in a small neighborhood of $Q = 1$. It is instructive to inquire how this loop changes when the complete equations of motion (25) are considered, rather than the Eqs. (20) which hold only near $Q = 1$. In effect, as the motion builds up the two curves of Fig. 3 cross as shown in Fig. 8 until, at the limit cycle, the areas enclosed by the positive and negative loops are equal.

Application

Here we shall apply the proposed mechanism of pulsating stars to a classical Cepheid variable. It is natural to choose δ-Cephei as a model because that star has been very extensively observed. We shall demonstrate that, when the numerical values of H_{e0}, H_{e2}, H_{c0}, H_{c2} and n are properly chosen, our system reproduces all essential features of the limit cycle of δ-Cephei.

Schwarzschild (1939) has calculated the radial velocity of the surface of δ-Cephei as a function of time, based on the observations of Danjon and others. Integration of this velocity-time curve gives the surface radius within a constant of integration. Schwarzschild has also computed this radius in terms of the mean radius as a function of time, but that mean radius was calculated by methods which have been much improved since Schwarzschild's work (1933). Therefore, we shall use the much more recent result of Oke (1961). Oke gives the value of $r/r_{\min}$ and of $\dot{r}$ as functions of the phase. These data must be converted to $Q = r/r_0$ and $dQ/d\tau$, where r_0 is the "equilibrium radius", before they can be compared to the results of our theory.

From the diagram in Fig. 2 we find that the apparent equilibrium radii for the expansion and compression phases are, respectively, $r_e/r_{\min} = 1.036$ and $r_c/r_{\min} = 1.030$. Hence, we chose the equilibrium radius as $r_0 = 1.033\, r_{\min}$.

From the transformation used to derive (11) one has

$$\frac{dQ}{d\tau} = \frac{1}{r_0}\left(\frac{GM}{r_0^3}\right)^{-1/2}\frac{dr}{dt}\,.$$

For the mean radius we use Oke's value

$$R = 40.3\, R_\odot = 28.0085 \times 10^{11}\ \text{cm}$$

where the subscript $\odot$ refers to the sun. This gives for the equilibrium radius

$$r_0 = 26.6585 \times 10^{11}\ \text{cm}.$$

To evaluate the time constant

$$\tau/t = (GM/r_0^3)^{-1/2}$$

one needs the mass of δ-Cephei. There is, however, some uncertainty as to that mass; for instance, Schwarzschild gave

$$M = 6.6\, M_\odot\,,$$

but Christy (1966) suggested recently a mass as low as

$$M = 3.3\, M_\odot\,.$$

We shall use neither of these values. Instead we utilize the relation between the period and the mean

Table 1

Phase	Q	$\dot{Q}$	Phase	Q	$\dot{Q}$
0.00	1.009	0.1691	0.50	1.093	−0.0520
0.05	1.031	0.1533	0.55	1.084	−0.0740
0.10	1.051	0.1303	0.60	1.072	−0.0962
0.15	1.068	0.1071	0.65	1.057	−0.1184
0.20	1.081	0.0838	0.70	1.039	−0.1407
0.25	1.091	0.0605	0.75	1.017	−0.1634
0.30	1.098	0.0374	0.80	0.993	−0.1778
0.35	1.102	0.0118	0.85	0.971	−0.0990
0.40	1.102	−0.0113	0.90	0.970	0.0589
0.45	1.099	−0.0316	0.95	0.985	0.1512

[2]) We owe this observation to a personal communication from Professor Ledoux. However, this acknowledgement should not be construed to imply that Professor Ledoux necessarily agrees with the content of this paper.

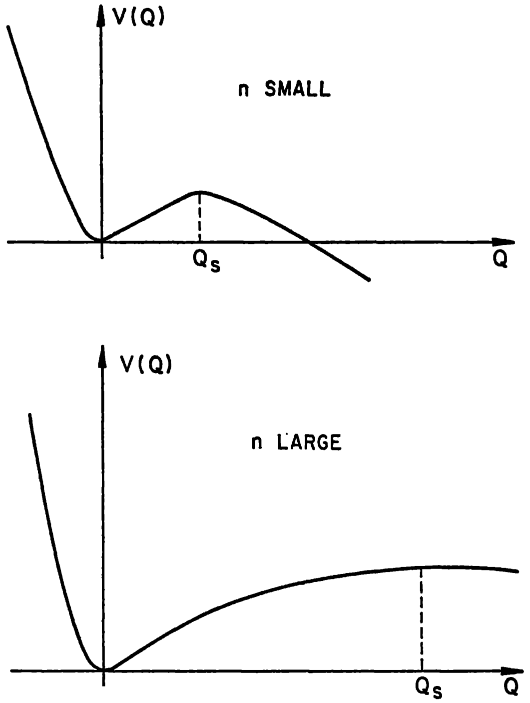

Fig. 9. Effect of small and large n on potential energy curves

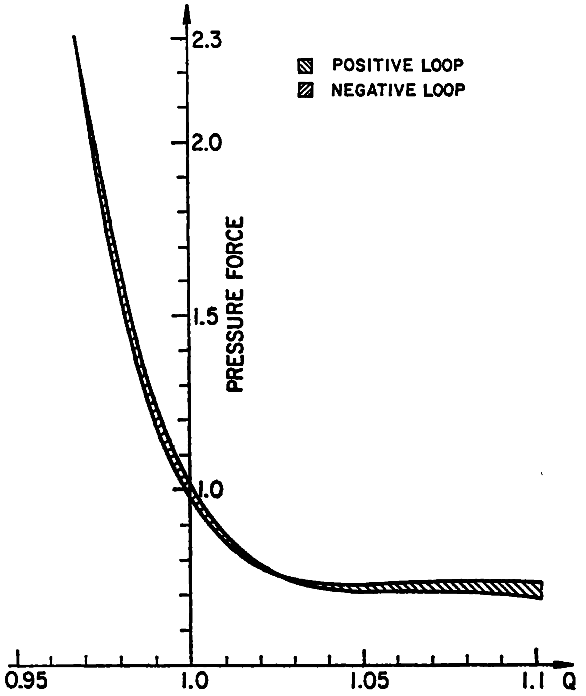

Fig. 10. Computed hysteresis loop of model of δ-Cephei at periodic motion

density, given by Kraft (1961) for classical Cepheids as

$$P\,(\varrho/\varrho_\odot) = 0.048$$

where P is the period in days. Since the period of δ-Cephei is 5.3663 days, one finds $\varrho/\varrho_\odot = 0.00008$, which gives the mass of δ-Cephei as

$$M = 5.236\, M_\odot .$$

Then, a simple calculation yields

$$(1/r_0)\,(GM/r_0^3)^{-1/2} = 0.0062\ (\mathrm{km/s})^{-1} .$$

This permits the conversion of Oke's table to our nondimensional quantities, and these are given in Table 1.

It is of interest to examine the effect of the magnitude of n on the shape of the potential energy curves (26), shown in Fig. 6. This effect is similar on, both the V_e and V_c-curves; hence, only one needs be examined.

The V-curves always have a minimum near $Q = 1$ when $n \geqq 3$; when H_{e2} (or H_{c2}) is zero, this is their only stationary point as shown in Fig. 5. However, when H_{e2} (or H_{c2}) is positive, these curves also have a maximum at some value $Q_s > 1$. The

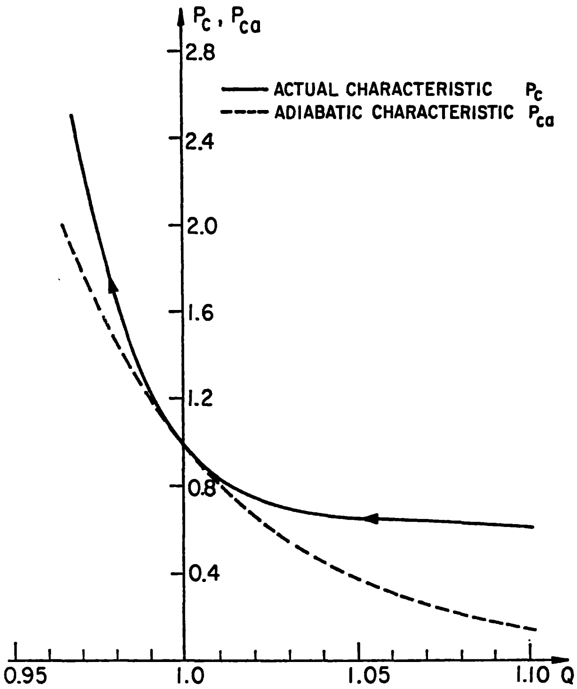

Fig. 11. The nonadiabatic effect on the pressure forces

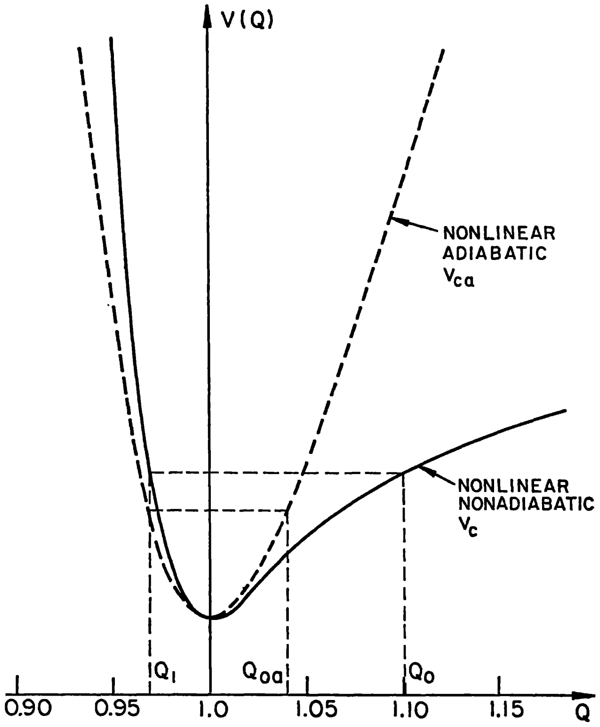

Fig. 12. The nonadiabatic effect on the potential energy curves and on symmetry of the motion

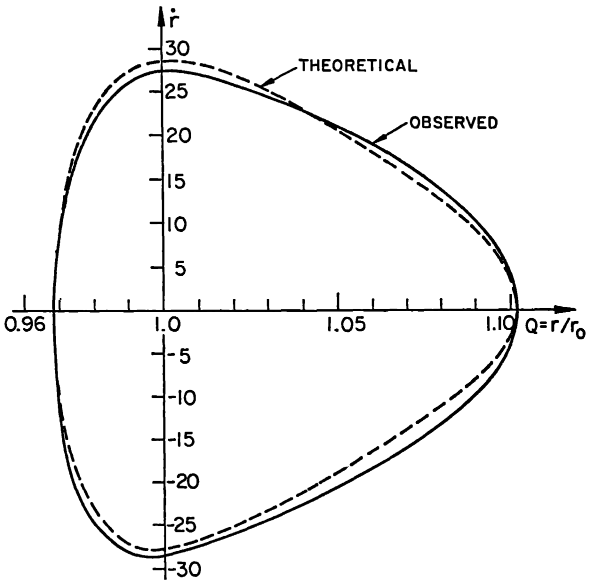

Fig. 13. Computed and observed limit cycle of δ-Cephei

effect of increasing n is one of increasing the value of Q_s as shown in Fig. 9.

To reproduce the features of the limit cycle of δ-Cephei we have chosen the numerical values given in Table 2.

These values satisfy the necessary conditions for the existence of a unique, stable limit cycle. The hysteresis loop for periodic motion may be calculated from the pressure forces

$$\hat{P}_e = \frac{H_{e0} + H_{e2}(1-Q)^2}{Q^n},$$

$$\hat{P}_c = \frac{H_{c0} + H_{c2}(1-Q)^2}{Q^n}.$$

Using the numerical values of Table 2, this hysteresis loop is shown in Fig. 10.

It is also instructive to identify certain terms in the equations of motion with adiabatic and non-adiabatic effects, and to study how the latter influence the pressure variation and the asymmetry of the motion discussed in connection with our second basic assumption.

Consider the pressure variation; during the compression phase the pressure is

$$P_c = [H_{c0} + H_{c2}(1-Q)^2]/Q^{n+2}$$

with a similar expression for the expansion phase. As mentioned in connection with (7), we regard as the adiabatic part the portion

$$P_{ca} = 1/Q^{n+2}.$$

The quantities P_{ca} and P_c are both plotted in Fig. 11 as functions of Q. It is seen that the major effect of the nonadiabetic term is to reduce the pressure variation in the domain $Q > 1$; i.e. on the expansion side of the equilibrium radius. This is also the result found by Usher and Whitney (1968) in the nonadiabatic case.

Similarly, one may examine the effect of the nonadiabatic terms on the asymmetry of the motion. For $n = 18$, the potential energy curve (for the compression phase) is

$$V_c = \frac{H_{c0} + H_{c2}}{17\,Q^{17}} - \frac{2H_{c2}}{16\,Q^{16}} + \frac{H_{c2}}{15\,Q^{15}} - \frac{1}{Q},$$

and the adiabatic component is

$$V_{ca} = \frac{1}{17\,Q^{17}} - \frac{1}{Q}.$$

Table 2

H_{e0}	H_{e2}	H_{c0}	H_{c2}	n
1.016	283.7	0.984	308.5	18

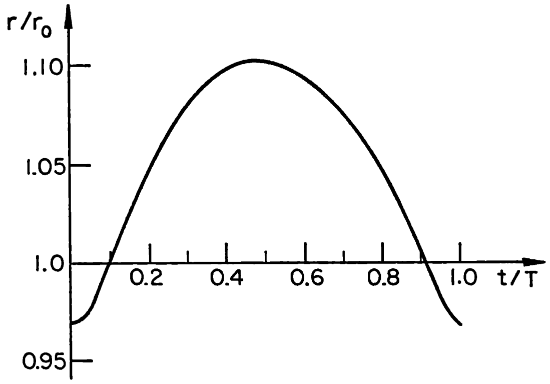

Fig. 14. Computed radial surface displacement of δ-Cephei

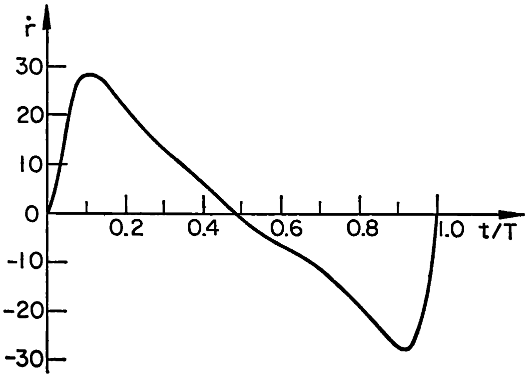

Fig. 15. Computed radial surface velocity of δ-Cephei

Table 3

Q	$\dot{Q}$	$\dot{r}$ km/s
0.968	0.000	0.00
0.975	0.1208	19.50
0.990	0.1717	27.71
1.001	0.1783	28.77
1.025	0.1615	26.06
1.060	0.1123	18.12
1.090	0.0571	9.21
1.102	0.0000	0.00
1.090	−0.0485	− 7.82
1.060	−0.0998	−16.10
1.025	−0.1518	−24.50
0.999	−0.1728	−27.88
0.990	−0.1680	−27.12
0.975	−0.1197	−19.31

Both, the V_{ca} and V_c-curves are plotted as functions of Q in Fig. 12. Also shown in Fig. 12 are the rest points Q_{0a} and Q_0 when $Q_1 = 0.968$ (the latter being Oke's result). One sees that the result of the non-adiabatic term is to produce a strong asymmetry $|Q_0| > |Q_1|$, whereas in the nonlinear, adiabatic case $|Q_{0a}| \cong |Q_1|$.

In Table 3 we give the theoretical values of Q and $\dot{Q}$ (and also of $\dot{r}$ in km/s) found from our theory, using the numerical values of Table 2

The quantities of Table 3 have been plotted in the phase plane and result in the dotted limit cycle-curve in Fig. 13. Also plotted in this diagram are Oke's values given in Table 1. They result in the solid limit cycle-curve of Fig. 13. It is seen that the theoretical curve reproduces quite faithfully the essential features of the observed limit cycle. Finally, theoretical results of r/r_0 and or $\dot{r}$ as functions of t/T (where T is the period) are shown in Figs. 14 and 15, respectively. A comparison with Fig. 1 shows that, again, the essential features of the observed values are preserved by the proposed mechanism.

References

Aleshin, V.I. 1959, *Sov. Astron.* **3**, 450.
Baker, N.H. 1966, Simplified Models for Cepheid Instability, Plenum Press, New York.
Baker, N.H., Moore, D.W., Spiegel, E.A. 1966, Abstr. in *Astron. Journ.* **71**, 844.
Baker, N.H., Moore, D.W., Spiegel, E.A. 1968, Further Studies of a Nonlinear Thermo-Mechanical Oscillator Rep. Goddard Space Flight Center, New York.
Chandrasekhar, S. 1939, An Introduction to the Study of Stellar Structure, Chicago University Press, Chicago.
Christy, R.F. 1966, *Astrophys. Journ.* **145**, 340.
Christy, R.F. 1968, *Quart. Journ. Roy. Astron. Soc.* **9**, 13.
Cox, J.P., Cox, A.N., Olsen, K.H., King, D.S., Eilers, D.D. 1966, *Astrophys. Journ.* **144**, 1038.
Kraft, R.P. 1961, *Astrophys. Journ.* **133**, 39.
Krogdahl, W.S. 1955, *Astrophys. Journ.* **122**, 43.
Ledoux, P. 1952, *Bull. Acad. Roy. Belg. Cl. Sci. Ser.* V, 38, 352.
Ledoux, P., Walraven, Th. 1958, *Handb. Phys.* **51**, 353.
Oke, J.B. 1961, *Astrophys. Journ.* **134**, 214.
Rauch, L.L. 1950, in *Contributions to the Theory of Nonlinear Oscillations*, ed. S. Lefschetz, Princeton University Press, Princeton.
Ritter, A. 1879, *Ann. d. Phys. u. Chem.* 8, 179.
Rudd, T.J. 1969, Ph. D. Dissertation, Univ. of Calif. Berkely.
Schwarzschild, M. 1939, Harvard Coll. Obs. Circular 431, Cambridge.
Usher, P.D., Whitney, C.A. 1968, *Astrophys. Journ.* **154**, 203.
Usher, P.D. 1968, Nonlinear Pulsation of Discrete Stellar Models, II, preprint.
Wesselink, A.J. 1939, *Astrophys. Journ.* **89**, 659.

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